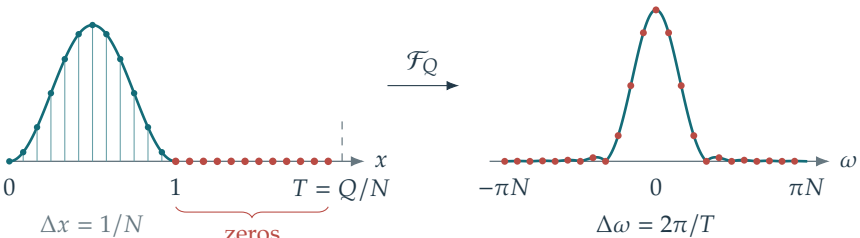
$f_n^Q = f(n/N), \quad N = 12, Q = 24$

continuous transform and DFT samples



$f(x) = \sin^2(\pi x)\,\mathbf{1}_{[0,1]}(x)$ $\quad |\widehat{f}(\omega)|$ $\quad |\widehat{f}_k^Q|/N$

$$\frac{1}{N}\widehat{f}_k^Q \approx \widehat{f}(2\pi k/T)$$